

2011 'A' Level H2 Mathematics Paper 2 9740/02
Section A: Pure Mathematics [40 marks]

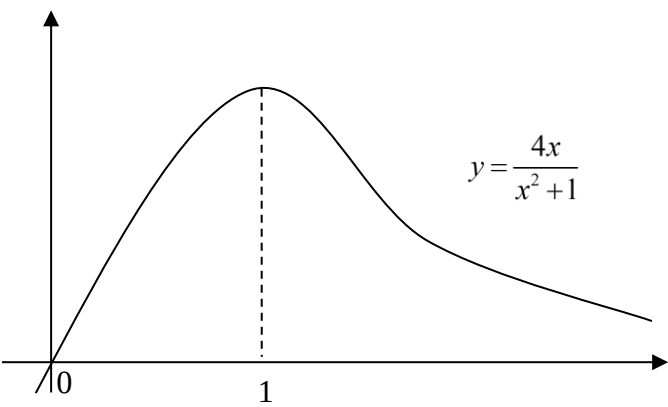
2	
(i)	Length = $2n - 2x$ Breadth = $n - 2x$ Height = x $\therefore$ Volume, $V = (2n - 2x)(n - 2x)x = 2n^2x - 6nx^2 + 4x^3$ (shown)
(ii)	$\frac{dV}{dx} = 2n^2 - 12nx + 12x^2$ n is a constant, given in the question. So $\frac{dn}{dx} = 0$. For stationary value of V, $\frac{dV}{dx} = 0$ $\Rightarrow 2n^2 - 12nx + 12x^2 = 0$ $\Rightarrow 6x^2 - 6nx + n^2 = 0$ $\Rightarrow x = \frac{6n \pm \sqrt{36n^2 - 4(6)(n^2)}}{12}$ May also use completing the square to get $6\left(x - \frac{n}{2}\right)^2 - \frac{n^2}{2}$. $x = \frac{6n \pm \sqrt{12n^2}}{12}$ Since $n \in \mathbb{Z}^+$, $x = \frac{6n \pm 2n\sqrt{3}}{12}$ $\therefore x = \frac{1}{6}(3 \pm \sqrt{3})n$ Note that Breadth = $n - 2x > 0 \Rightarrow x < \frac{n}{2}$ $x = \frac{n}{2} + \frac{\sqrt{3}}{6}n > \frac{n}{2}$. Hence, rejected Thus $x = \frac{1}{6}(3 + \sqrt{3})n$ is rejected. $\therefore x$ must be $\frac{1}{6}(3 - \sqrt{3})n$, because x must be less than $\frac{n}{2}$.

Answer must be in simplified form, including the surd (to be rationalised).

Write down your reason.

3	
(i)	Let $y = \ln(2x+1)+3$. $2x+1 = e^{y-3}$ $x = \frac{e^{y-3}-1}{2}$ $\therefore f^{-1}(x) = \frac{e^{x-3}-1}{2}$ $D_{f^{-1}} = R_f = \mathbb{R}, \quad R_{f^{-1}} = D_f = \left(-\frac{1}{2}, \infty\right)$
(ii)	The functions are still increasing, so do not 'bend back on itself'. Question asks for 'exact coordinates'.
(iii)	The points of intersection also lies on the line $y = x$, thus the x-coordinates also satisfy the equation of $f(x) = x \Rightarrow \ln(2x+1)+3 = x \Rightarrow \ln(2x+1) = x-3$. By GC, $x = -0.4847, 5.482$ (4 s.f.)

There are 2 values of x so make sure the graphs in part (ii) intersect at 2 points. You may need to adjust the window setting in your G.C. in order to see clearly.

Q4	Suggested Solutions
(ai)	$\int_0^n x^2 e^{-2x} dx = \left[x^2 \left(-\frac{1}{2} e^{-2x} \right) \right]_0^n - \int_0^n (2x) \left(-\frac{1}{2} e^{-2x} \right) dx$ $= -\frac{1}{2} n^2 e^{-2n} + \int_0^n x e^{-2x} dx$ $= -\frac{1}{2} n^2 e^{-2n} + \left[x \left(-\frac{1}{2} e^{-2x} \right) \right]_0^n - \int_0^n -\frac{1}{2} e^{-2x} dx$ $= -\frac{1}{2} n^2 e^{-2n} - \frac{1}{2} n e^{-2n} + \frac{1}{2} \left[-\frac{1}{2} e^{-2x} \right]_0^n$ $= -\frac{1}{2} n^2 e^{-2n} - \frac{1}{2} n e^{-2n} - \frac{1}{4} e^{-2n} + \frac{1}{4}$ $= \frac{1}{4} - \left(\frac{n^2}{2} + \frac{n}{2} + \frac{1}{4} \right) e^{-2n}$
(aii)	As $n \rightarrow \infty$, $n e^{-2n} \rightarrow 0$, $n^2 e^{-2n} \rightarrow 0$ and $e^{-2n} \rightarrow 0$, so $\int_0^\infty x^2 e^{-2x} dx = 0 + 0 + 0 + \frac{1}{4}$ $= \frac{1}{4}$ Must use result in part (i) due to 'Hence' being used in question.
(b)	 $y = \frac{4x}{x^2 + 1}$ $\text{Volume} = \pi \int_0^1 y^2 dx$ $= \pi \int_0^1 \left(\frac{4x}{x^2 + 1} \right)^2 dx$ $= \pi \int_0^1 \frac{16x^2}{(x^2 + 1)^2} dx$ When $x = 0$, $\theta = 0$ When $x = 1$, $\theta = \frac{\pi}{4}$

$x = \tan \theta$ $\frac{dx}{d\theta} = \sec^2 \theta$ $\therefore \pi \int_0^1 \frac{16x^2}{(x^2+1)^2} dx = \pi \int_0^{\frac{\pi}{4}} \frac{16 \tan^2 \theta}{(\tan^2 \theta + 1)^2} \sec^2 \theta d\theta$ $= \pi \int_0^{\frac{\pi}{4}} \frac{16 \tan^2 \theta}{(\sec^2 \theta)^2} \sec^2 \theta d\theta$ $= \pi \int_0^{\frac{\pi}{4}} \frac{16 \tan^2 \theta}{\sec^2 \theta} d\theta$ $= 16\pi \int_0^{\frac{\pi}{4}} \sin^2 \theta d\theta$ $= 16\pi \int_0^{\frac{\pi}{4}} \frac{1}{2} (1 - \cos 2\theta) d\theta$ $= 8\pi \left[\theta - \frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{4}}$ $= 8\pi \left(\frac{\pi}{4} - \frac{1}{2} \right)$ $= 2\pi^2 - 4\pi = 2\pi(\pi - 2)$	Must only have 'θ' in this expression. Do not mix 'x' with 'θ' here.
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Section B: Statistics [60 marks]

Qn 5	Suggested Solutions
	$X \sim N(\mu, \sigma^2)$ $P(X < 70.0) = 0.975$ $P\left(Z < \frac{70 - \mu}{\sigma}\right) = 0.975$ $\frac{70 - \mu}{\sigma} = 1.96$ $1.96\sigma + \mu = 70 \text{ --- (1)}$ $P(X < 40.0) = 0.05$ $P\left(Z < \frac{40 - \mu}{\sigma}\right) = 0.05$ $\frac{40 - \mu}{\sigma} = -1.6449$ $-1.6449\sigma + \mu = 40 \text{ --- (2)}$ Using GC, solving equation (1) and (2) $\sigma = 8.3220 \approx 8.32$ $\mu = 53.689 \approx 53.7$

7	Suggested Solutions
(i)	(a) The probability of success in contacting a friend remains constant at 0.7 throughout the sample. (b) The success of each phone call is independent of the success of other phone calls.
(ii)	The probability of success in contacting a friend may not be constant for different people. Alternatively, the success of a phone call may not be independent of the success of other phone calls. For example, knowing one of his friends cannot be contacted will lead to knowing another friend is not contactable.
(iii)	$R \sim B(8, 0.7)$ $P(R \geq 6) = 1 - P(R \leq 5)$ $= 1 - 0.44823$ $= 0.55177$ $= 0.552$

Q8	Suggested Solutions
(i)	
	GC, $r = -0.992$ (3 s.f.) (2.0, 18.8) (5.0, 0.8) Remember to label the first and last points on the scatter diagram. Even though r is close -1, from the scatter diagram, as x increases, y decreases at an increasing rate, hence a quadratic model might be more suitable.
(iii)	Let r_1 and r_2 denotes the product moment correlation coefficients based on the linear model and the quadratic model respectively. If $r_2 > r_1$, then we would deem the quadratic model to be the better model. If $r_2 < r_1$, then the linear model would be better.

	OR Since $r_2 = -0.99998 = 0.99998$ is closer to 1 as compared to $r_1 = -0.99232 = 0.99232$, the quadratic model is more suitable.
(iv)	Using GC, the equation of the least-squares regression line of y on x^2 is $y = -0.85621x^2 + 22.23049$ $y = -0.856x^2 + 22.2 \text{ (3 s.f.)}$ When $x = 3.2$, the estimate value of y is $y = -0.85621(3.2)^2 + 22.23049 = 13.5 \text{ (3 s.f.)}$ Give the values of a and b to at least 5 s.f. first as these will be used to estimate the value of y when $x = 3.2$.

9	Suggested Solutions
(i)	Let A denotes the event that “the selected lens is made by A”. Let B denotes the event that “the selected lens is made by B”. Let F denotes the event that “the selected lens faulty”. Let F' denotes the event that “the selected lens is not faulty”. <pre> graph LR Root(()) --- 0.6 A((A)) Root --- 0.4 B((B)) A --- 0.05 F1((F)) A --- 0.95 Fp1((F')) B --- 0.07 F2((F)) B --- 0.93 Fp2((F')) </pre>
(i)(a)	$P(F) = 0.6(0.05) + 0.4(0.07) = 0.058$
(i)(b)	$P(A F) = \frac{P(A \cap F)}{P(F)} = \frac{0.05 \times 0.6}{0.058} = 0.517 \text{ (to 3 s.f.)}$
(ii)(a)	$P(\text{exactly one is faulty}) = 0.058 \times 0.942 \times 2!$ $= 0.109272 \text{ (exact)}$
(ii)(b)	$P(\text{both are made by } A \mid \text{exactly one is faulty})$ $= \frac{P(\text{both are made by } A \text{ and exactly one is faulty})}{P(\text{exactly one is faulty})}$ $= \frac{0.6 \times 0.05 \times 0.6 \times 0.95 \times 2!}{0.109272}$ $= 0.313 \text{ (to 3s.f.)}$ Multiply by $2!$ due to permutation.

Qn 10	Solution
(i)	Let μ denote the population mean time taken to install a component. $H_0: \mu = 38.0$ $H_1: \mu < 38.0$ Must always remember to define μ.
(ii)	Given T is the time taken for an employee to install an electronic component. Let σ denote the population standard deviation. Under H_0, Given $T \sim N(38.0, 5.0^2)$, $\therefore \bar{T} \sim N\left(38.0, \frac{5.0^2}{50}\right)$ Test Statistic: $Z = \frac{\bar{T} - 38.0}{5/\sqrt{50}}$ Level of significance: 5% Reject H_0 if $p\text{-value} < 0.05$ i.e. For H_0 to be rejected, $z\text{-value} < -1.6449$ $z\text{-value} = \frac{\bar{t} - 38.0}{5/\sqrt{50}} < -1.6449$ $\Rightarrow \bar{t} < 36.837$ Therefore, the set of values is $\{\bar{t} \in \mathbb{R} : 0 < \bar{t} < 36.8\}$.
(iii)	$H_0: \mu = 38.0$ $H_1: \mu < 38.0$ Under H_0, Given $T \sim N(38.0, 5.0^2)$, $\therefore \bar{T} \sim N\left(38.0, \frac{5.0^2}{50}\right)$ Test Statistic: $Z = \frac{\bar{T} - 38.0}{5/\sqrt{50}}$ Level of significance: 5% Reject H_0 if $p\text{-value} < 0.05$ i.e. For H_0 not to be rejected, $z\text{-value} > -1.6449$ $z\text{-value} = \frac{37.1 - 38.0}{5/\sqrt{n}} > -1.6449$ $\Rightarrow \frac{-0.9}{5} \sqrt{n} > -1.6449$ $\Rightarrow n < \left[(-1.6449) \left(\frac{5}{0.9} \right) \right]^2 = 83.504$ Therefore, the set of values is $\{n \in \mathbb{Z}^+ : n \leq 83\}$ OR $\{n \in \mathbb{Z} : 0 < n \leq 83\}$ Remember to change inequality sign when multiplied by negative number. Answer must be whole number since n is an integer.

Q11	Suggested Solutions
(i)	$P(R=4) = \frac{\binom{18}{4} \binom{12}{6}}{\binom{30}{10}} = 0.0941$
(ii)	$P(R=r) > P(R=r+1)$ $\frac{\binom{18}{r} \binom{12}{10-r}}{\binom{30}{10}} > \frac{\binom{18}{r+1} \binom{12}{9-r}}{\binom{30}{10}}$ Refer to MF26 for the formula of $\binom{n}{x}$. $\frac{\binom{18}{r} \binom{12}{10-r}}{\binom{30}{10}} > \frac{\binom{18}{r+1} \binom{12}{9-r}}{\binom{30}{10}}$ $\frac{18!}{r!(18-r)!(10-r)!(2+r)!} > \frac{18!}{(r+1)!(17-r)!(9-r)!(3+r)!}$ $r!(18-r)!(10-r)!(2+r)! < (r+1)!(17-r)!(9-r)!(3+r)!$ $(r+1)!(17-r)!(9-r)!(r+3)! > r!(18-r)!(10-r)!(r+2)! \quad (\text{shown})$ $(r+1)(r+3) > (18-r)(10-r)$ $r^2 + 4r + 3 > 180 - 28r + r^2$ $32r > 177$ $r > 5.53125$ $\therefore r = 6$ Note: You may use GC to solve for r but working must be shown clearly.